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Lesson 4: Classification and Evaluation Metrics

Technologies for Artificial Intelligence

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Before we start

- Exercise 3, which we discuss now
- Ridge, Lasso, and the choice of the penalty strength

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Today: from a number to a decision

- Logistic regression: the model and why the cost had to change
- The confusion matrix, and why accuracy lies
- Precision, recall, F1, and the threshold nobody chose
- Receiver operating characteristic (ROC) curves, and where they mislead
- Choosing a threshold from money

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The question has changed

- Lesson 3: given a house, **how many euros?**
- Lesson 4: given a disk drive, **will it fail?**
- Answer with a probability: $0 \leq P(y = 1 \mid x) \leq 1$
- $y = 1$ if the drive failed, $y = 0$ if not; x holds its six counters

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The fleet

- **8,000 disk drives** in a data centre
- Six SMART counters each: Self-Monitoring, Analysis and Reporting Technology
- Reallocated sectors, spin retries, read and seek error rates, temperature, hours in service
- Label: did it fail within thirty days?
- `train_test_split(..., stratify=y)`: 6,000 to train, 2,000 to test

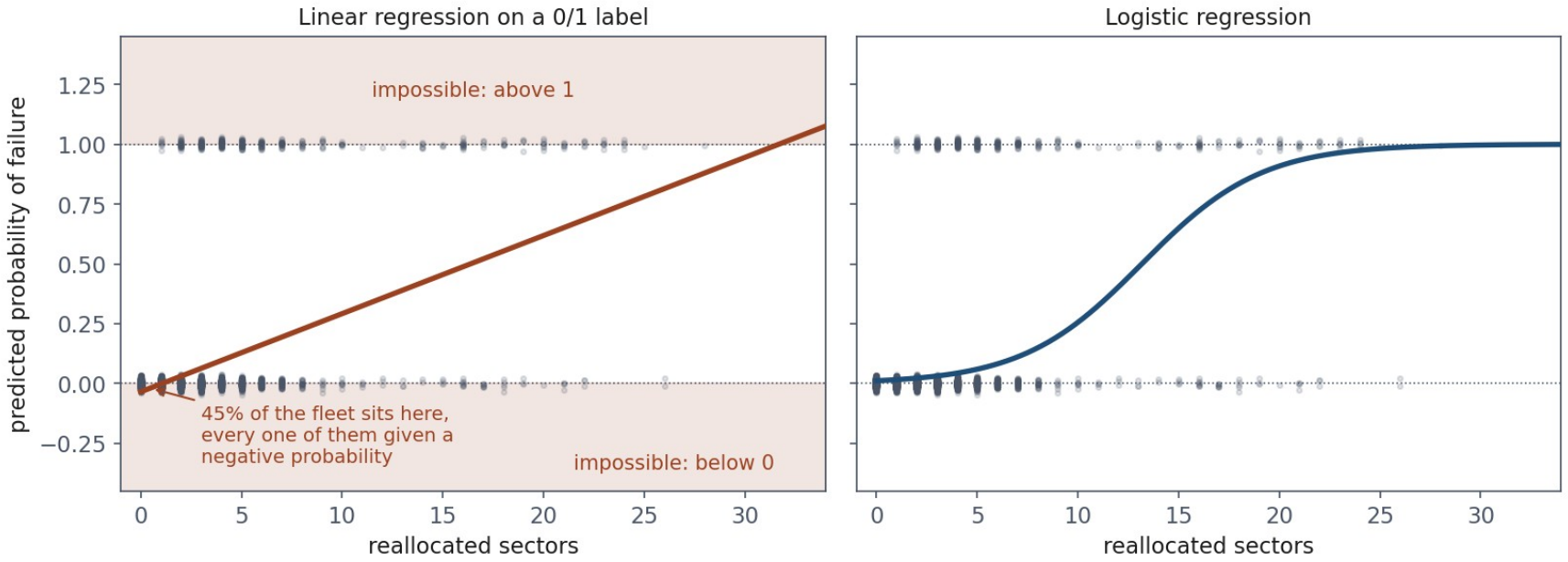
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306

- **306 of the 8,000 drives failed**: 306 ÷ 8,000 = 3.8% of the fleet
- Which means 100% − 3.8% = 96.2% did not

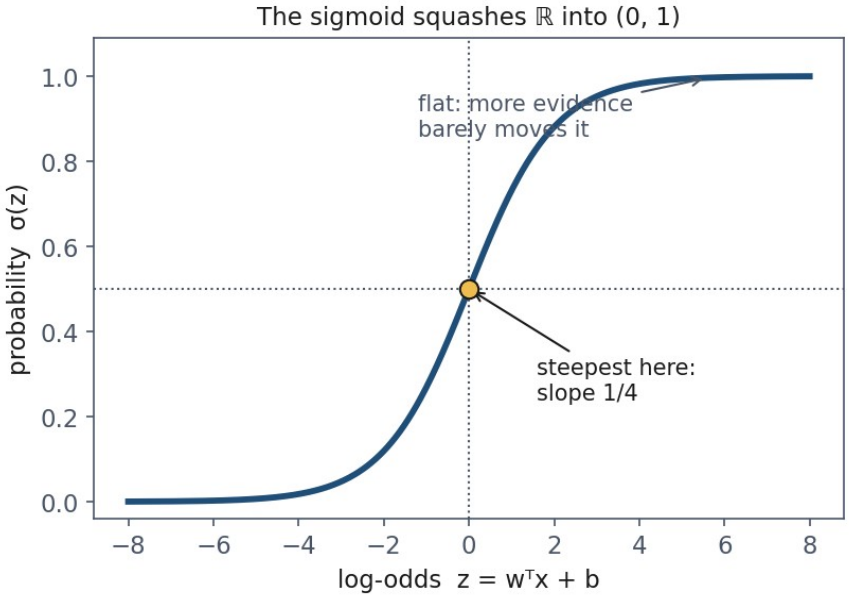
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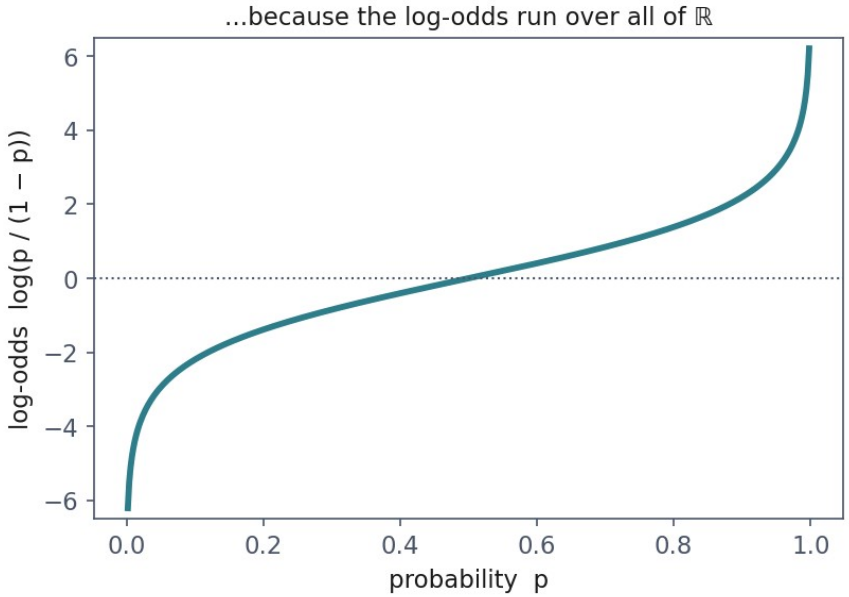
What goes wrong if we just fit a line



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The sigmoid





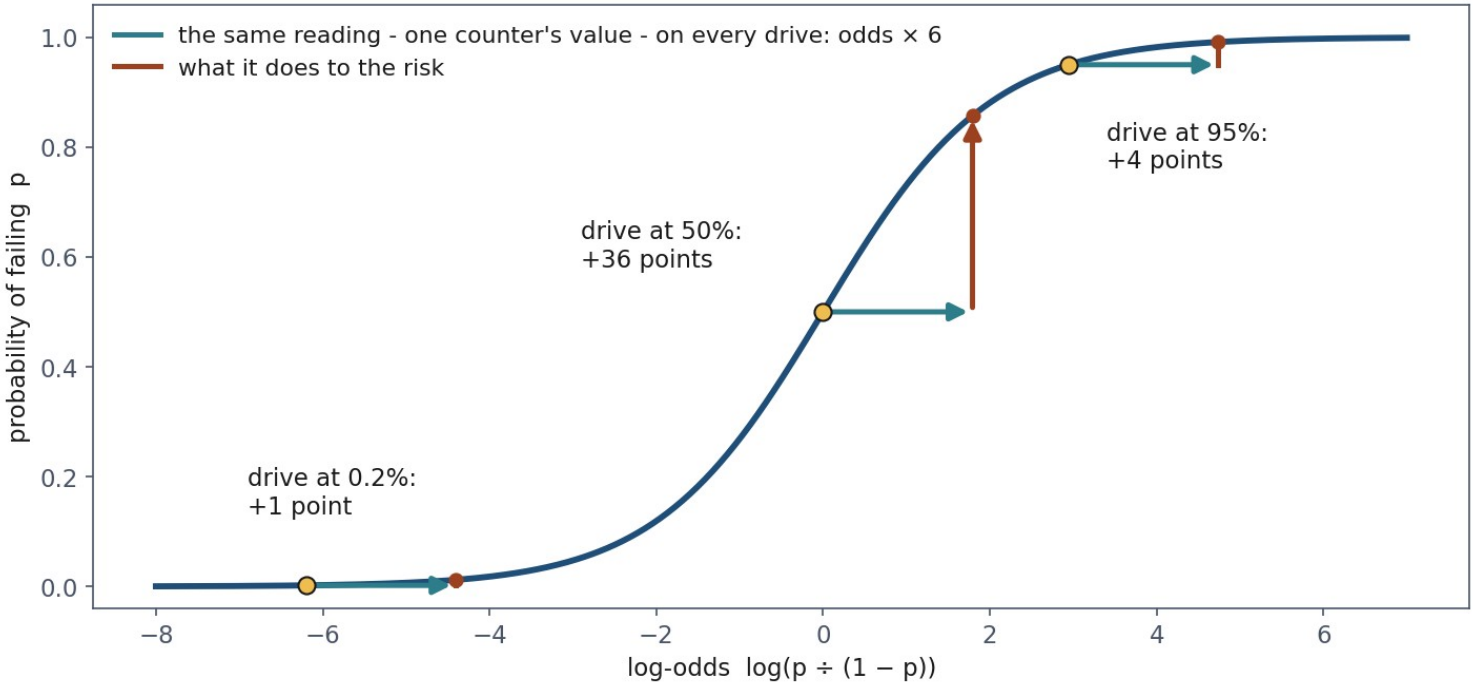
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Why the log-odds

- A line gives any number; a probability, only [0, 1]
- Odds, $p \div (1 - p)$: no ceiling, from 0 to ∞
- Log of the odds: no floor either, $-\infty$ to $+\infty$
- $p = 0.99$: $\log(0.99 \div 0.01) = \log 99 = +4.60$; $p = 0.5$: $\log(0.5 \div 0.5) = \log 1 = 0$
- Bonus: on this scale each feature will add a fixed amount, so it will multiply the odds by a fixed factor (slide 14)

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One reading, three different jumps



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Log-odds of p = P(y = 1 | x): a line

$$\log \frac{p}{1-p} = w^\top x + b \quad \Longrightarrow \quad p = \frac{1}{1 + e^{-(w^\top x + b)}}$$

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The model, and three properties of the sigmoid

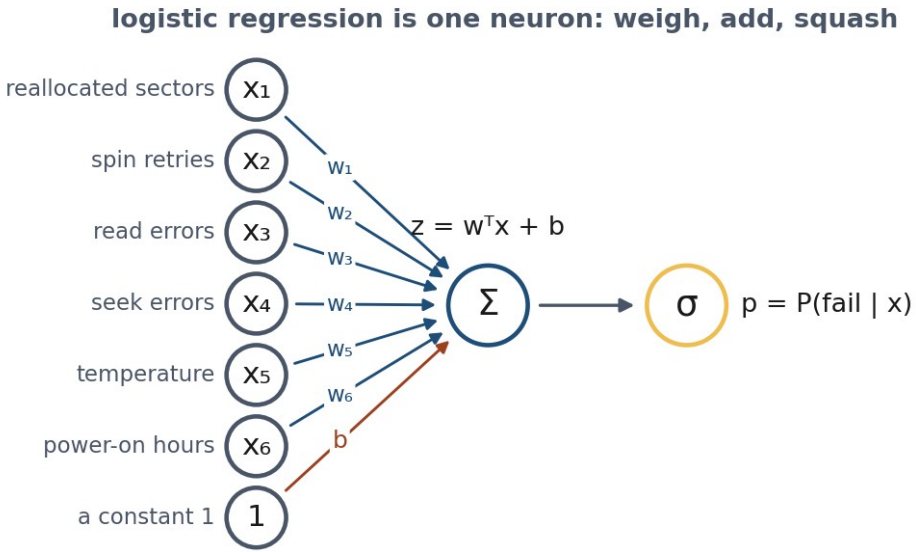
$$P(y = 1 \mid x) = \sigma(w^\top x + b), \qquad \sigma(z) = \frac{1}{1 + e^{-z}}$$

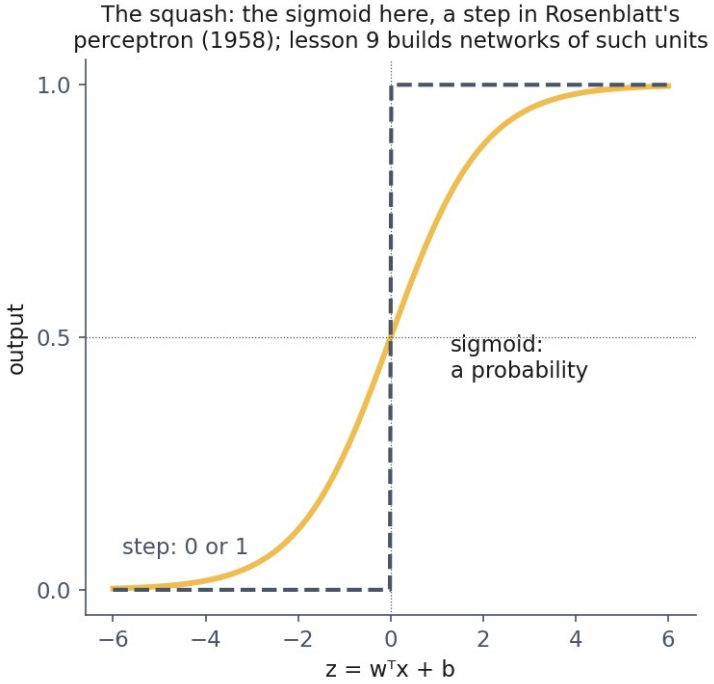
$$\sigma(0) = \frac{1}{2}, \qquad \sigma(-z) = 1 - \sigma(z)$$

$$\sigma'(z) = \sigma(z)\,(1 - \sigma(z)) \leq \frac{1}{4}$$

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The model, drawn: one neuron





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What the coefficients mean

- Coefficients = weights w (`coef_`); intercept b (`intercept_`)
- Standardised (mean 0, SD 1): one SD more of reallocated sectors adds its true coefficient, 1.80, to the log-odds
- log(odds) + 1.80 = log(odds × 6.05): **six times the odds**, slide 10's teal step
- In code: `make_pipeline(StandardScaler(), LogisticRegression())`

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The truth, and what we recovered

Feature	True, `disk_data.py`	Fitted, notebook 1	Odds ×, exp(Fitted)
reallocated_sectors	1.80	1.66	5.28
spin_retry_count	1.15	1.17	3.22
read_error_rate	0.85	0.87	2.39
temperature_c	0.45	0.46	1.59
power_on_hours	0.35	0.38	1.46
seek_error_rate	**0.00**	**0.02**	**1.02**

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The intercept is the average drive’s chance

- Fitted intercept: **−6.09**, the log-odds of a drive average on every counter (all standardised x = 0)
- $\sigma(-6.09) = 1 \div (1 + \exp(6.09)) = 1 \div 442 = 0.0023$: a 0.2% chance
- The fleet fails at 3.8% (slide 6): the failures come from the few drives far above average

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Fitting means minimising something, but what?

- Squared error worked for lesson 3
- Why not just apply it to the sigmoid’s output?

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Step 1: each drive gets a p

Drive	What happened	y	Setting 1: $p = \sigma(w^T x + b)$, the P(fail)
A	failed	1	0.8
B	stayed healthy	0	0.1
C	stayed healthy	0	0.2

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Step 2: how likely was the outcome?

Drive	y	p	Probability of what happened: p if y = 1, 1 - p if y = 0
A	1	0.8	0.8
B	0	0.1	$1 - 0.1 = 0.9$
C	0	0.2	$1 - 0.2 = 0.8$
All three together, if the drives are independent: multiply, as for two coins (½ × ½ = ¼)			$0.8 \times 0.9 \times 0.8 =$ **0.576**

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Step 2 as formulas: one drive, then the likelihood

$$P(y \mid x) = p^{\,y}\,(1-p)^{1-y}$$

$$\mathcal{L}(w,b) = \prod_{i=1}^{m} \left(p^{(i)}\right)^{y^{(i)}} \left(1-p^{(i)}\right)^{1-y^{(i)}}$$

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Step 3: two settings, the likelier wins

Drive	y	Setting 1: p	Setting 1: probability of what happened	Setting 2: p	Setting 2: probability of what happened
A	1	0.8	0.8	0.3	0.3
B	0	0.1	$1 - 0.1 = 0.9$	0.6	$1 - 0.6 = 0.4$
C	0	0.2	$1 - 0.2 = 0.8$	0.5	$1 - 0.5 = 0.5$
Likelihood: the product			$0.8 \times 0.9 \times 0.8 =$ **0.576**		$0.3 \times 0.4 \times 0.5 =$ **0.060**

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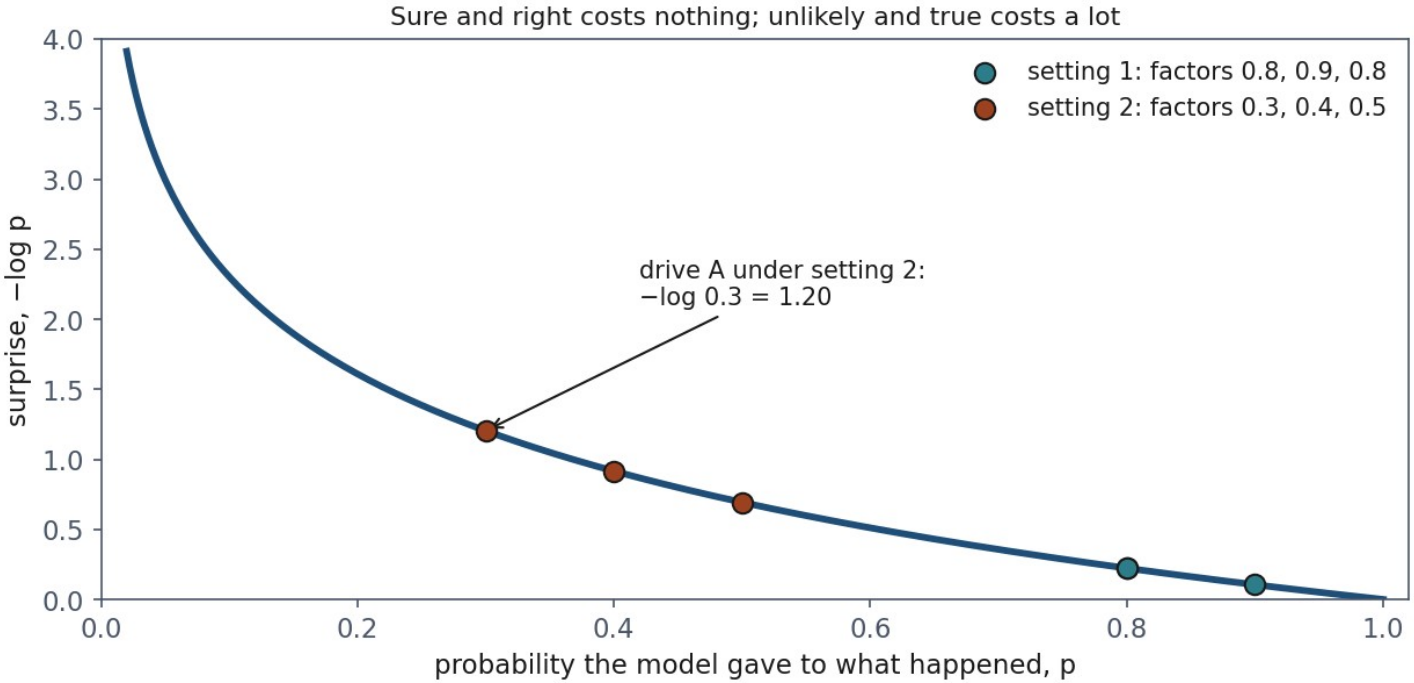
Step 4: minus the log, a cost

The log turns the product into a sum, one term per drive, and keeps the winner; the minus turns largest into smallest: a cost to minimise, as in lesson 3.

Drive	y	Setting 1: −log of probability of what happened	Setting 2: −log of probability of what happened
A	1	−log 0.8 = 0.22	−log 0.3 = 1.20
B	0	−log 0.9 = 0.11	−log 0.4 = 0.92
C	0	−log 0.8 = 0.22	−log 0.5 = 0.69
Sum = −log of the likelihood		0.22 + 0.11 + 0.22 = **0.55**	1.20 + 0.92 + 0.69 = **2.81**

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Surprise: minus the log of a probability



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Cross-entropy, also called log loss

$$L = -[\, y \log p + (1-y)\log(1-p)]$$

$$J(w,b) = \frac{1}{m}\sum_{i=1}^{m} L^{(i)}, \qquad p^{(i)} = \sigma(w^{\top} x^{(i)} + b)$$

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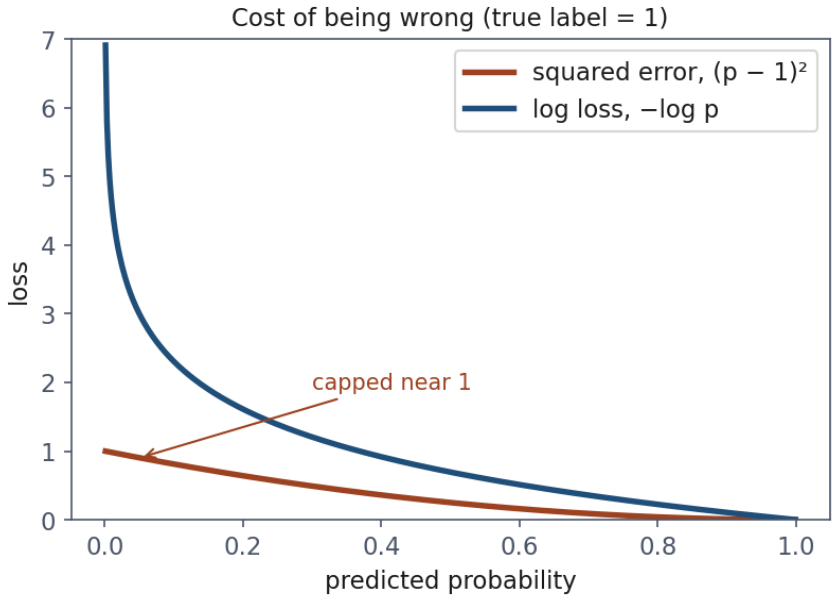
Lesson 3’s gradient, $\hat{y} = \sigma(w^T x + b)$

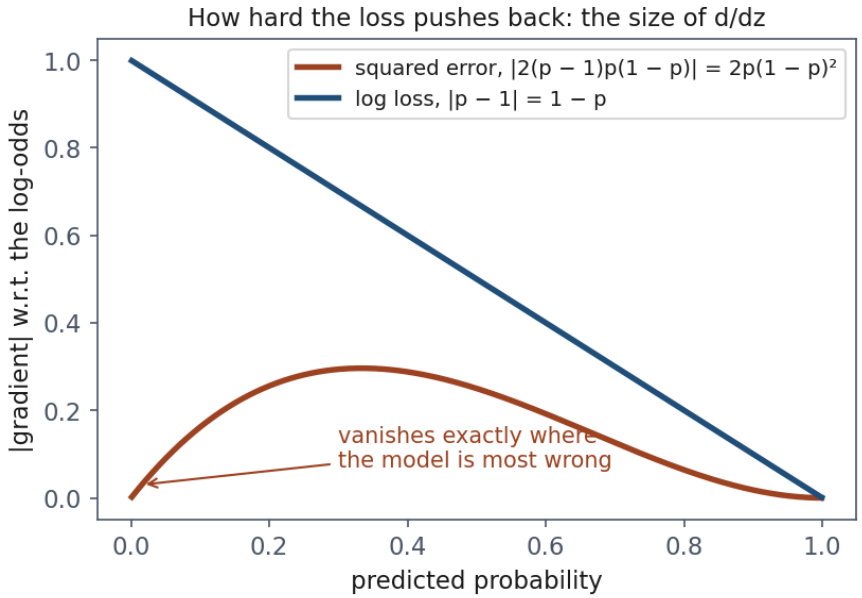
$$\frac{\partial J}{\partial w_j} = \frac{1}{m} \sum_{i=1}^{m} \left(\hat{y}^{(i)} - y^{(i)} \right) x_j^{(i)}$$

$$\frac{\partial J}{\partial b} = \frac{1}{m} \sum_{i=1}^{m} \left(\hat{y}^{(i)} - y^{(i)} \right)$$

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So why not squared error?





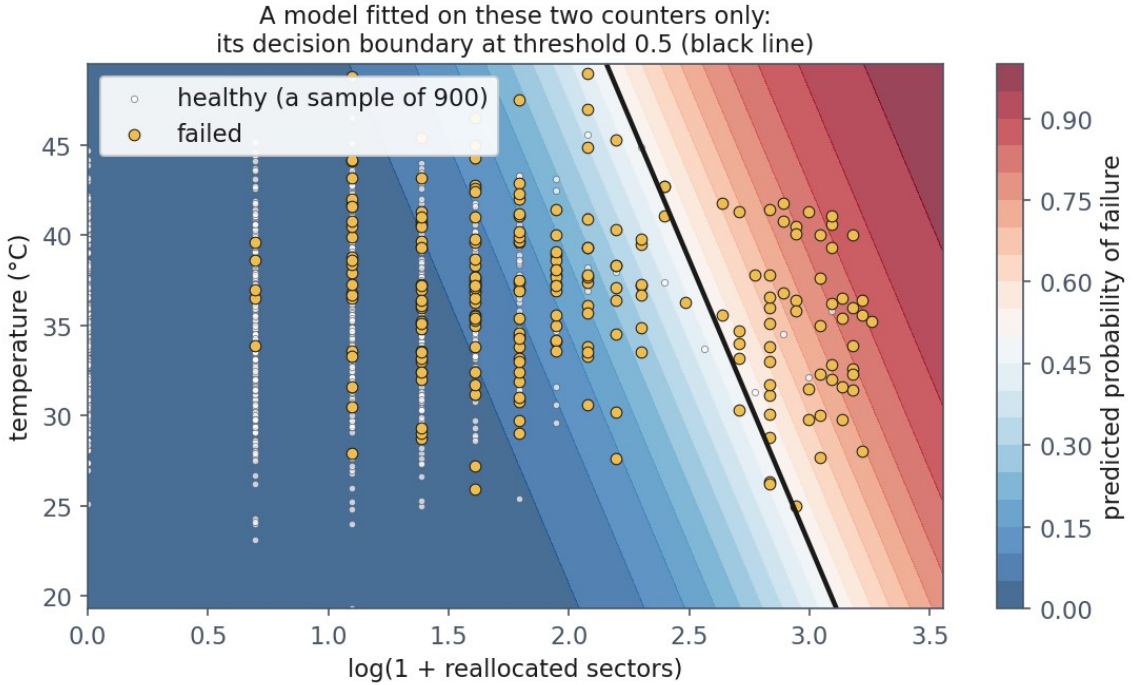
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And it is convex, which squared error is not

- Log loss is **convex** in the weights, so any local minimum is global
- Squared error through a sigmoid is not
- Why not, from slide 26: its slope falls from 0, then climbs back to 0; a convex curve’s slope only grows

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The decision boundary of a two-counter model



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Notebook 1, live

section, and its slides	you call (→ the number to come back with)
a line on a 0/1 label, 7	`LinearRegression` → 3,606 negative probabilities
odds, and why not squared error, 8-27	NumPy only: the sigmoid, the two losses, their gradients
gradient descent, 25	`train_test_split`, `StandardScaler`, fifteen lines of NumPy → log loss 0.6931 to 0.0692
the decision boundary, 28	`make_pipeline(StandardScaler(), LogisticRegression())` on two features
the truth, recovered, 15-16	`LogisticRegression`, `.coef_`, `.intercept_` → seek_error_rate 0.02, intercept −6.09

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Break

- Ten minutes

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Two models

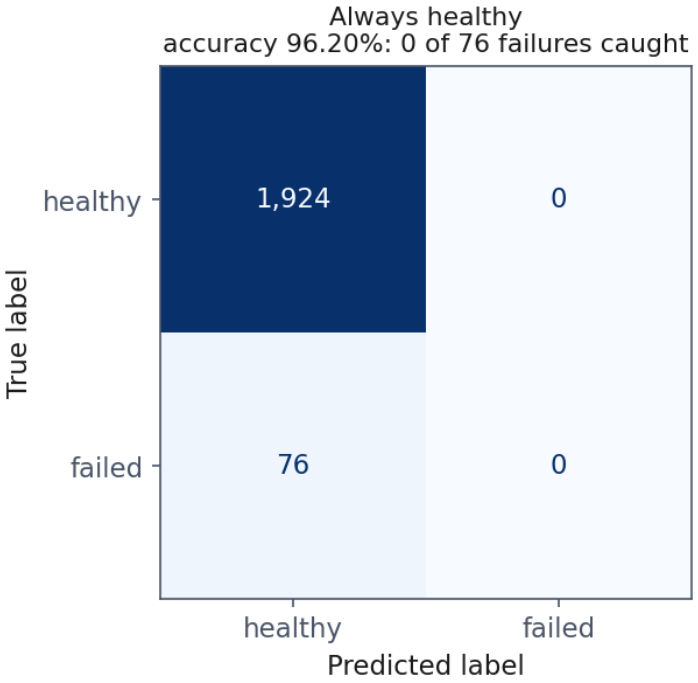
Model, on the 2,000 test drives (76 fail)	Accuracy = right ÷ 2,000	Failures caught
Predict “healthy” for everything	every healthy drive: 1,924 ÷ 2,000 = **96.20%**	**0 of 76**
Logistic regression, threshold 0.5	43 failures + 1,911 healthy: 1,954 ÷ 2,000 = 97.70%	43 of 76

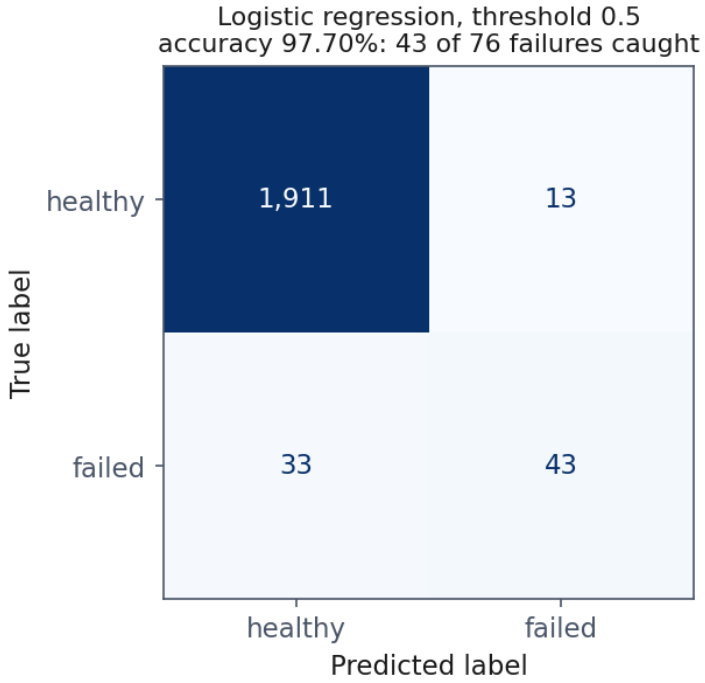
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Why everyone makes this mistake once

- Accuracy is the fraction of predictions that were right
- On a **balanced** problem whose two errors cost the same, that is exactly what you want
- The interesting class is almost always the **rare** one

Four numbers instead of one





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The four outcomes, and what they cost

- **False positive (FP)**: we replace a healthy drive. A technician’s hour, €140
- **False negative (FN)**: a drive dies in service. Callout, rebuild, risk, €2,600
- Both costs are assumptions supplied from outside the data, not measured
- **True positive (TP)** and **true negative (TN)**, the right calls, count as €0: each cost is what the error adds

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Precision and recall: look at the denominator

Precision = TP / (TP + FP)
"of the alarms we raised"

	predicted healthy	predicted failing
actually healthy	TN 1911	**FP** 13
actually failed	FN 33	**TP** 43

Recall = TP / (TP + FN)
"of the drives that failed"

	predicted healthy	predicted failing
actually healthy	TN 1911	FP 13
actually failed	**FN** 33	**TP** 43

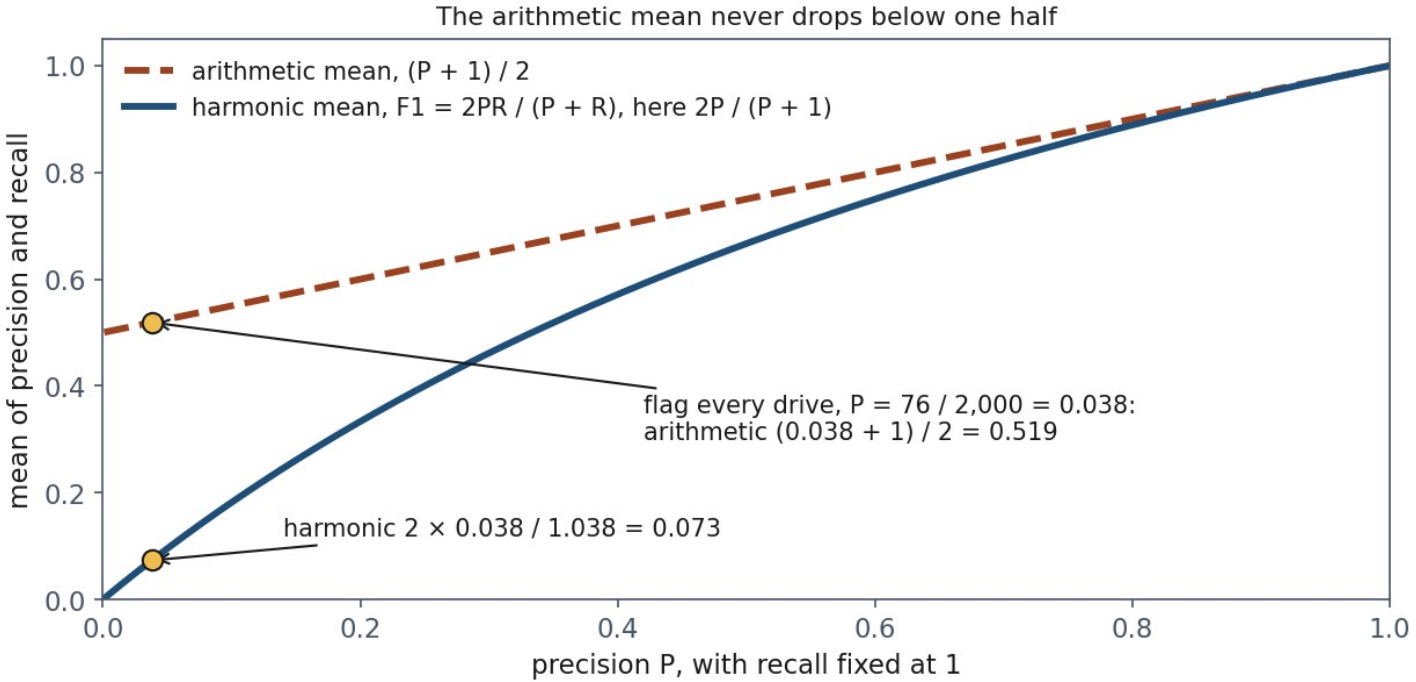
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Our model, in those terms

- True and false positives and negatives: TP, FP, FN, TN
- Precision = TP ÷ (TP + FP) = 43 ÷ (43 + 13) = **0.768**: of the drives flagged
- Recall = TP ÷ (TP + FN) = 43 ÷ (43 + 33) = **0.566**: of the drives that failed
- Specificity = TN ÷ (TN + FP) = 1,911 ÷ 1,924 = 0.993; 1 − 0.993 is the false positive rate (FPR)
- In code: `precision_score`, `recall_score`

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F1 = 2PR ÷ (P + R): why the harmonic mean



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Read the minority class, not the average

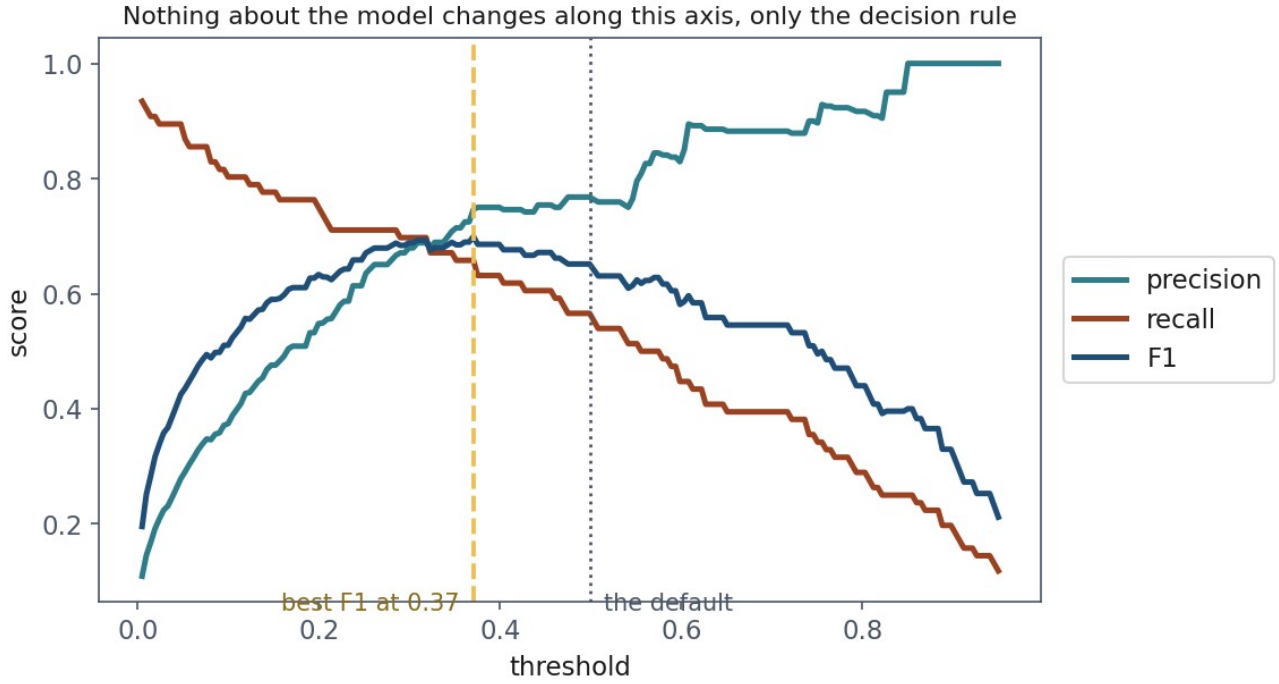
- F1 per class: healthy 0.988 (1,924 drives), failed 0.652 (76 drives)
- `macro avg` F1 = (0.988 + 0.652) ÷ 2 = **0.820**: both classes count the same
- `weighted avg` F1 = (1,924 × 0.988 + 76 × 0.652) ÷ 2,000 = **0.975**: the healthy class dominates
- In code: `classification_report(y_test, y_pred)` prints both rows

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Notebook 2, live

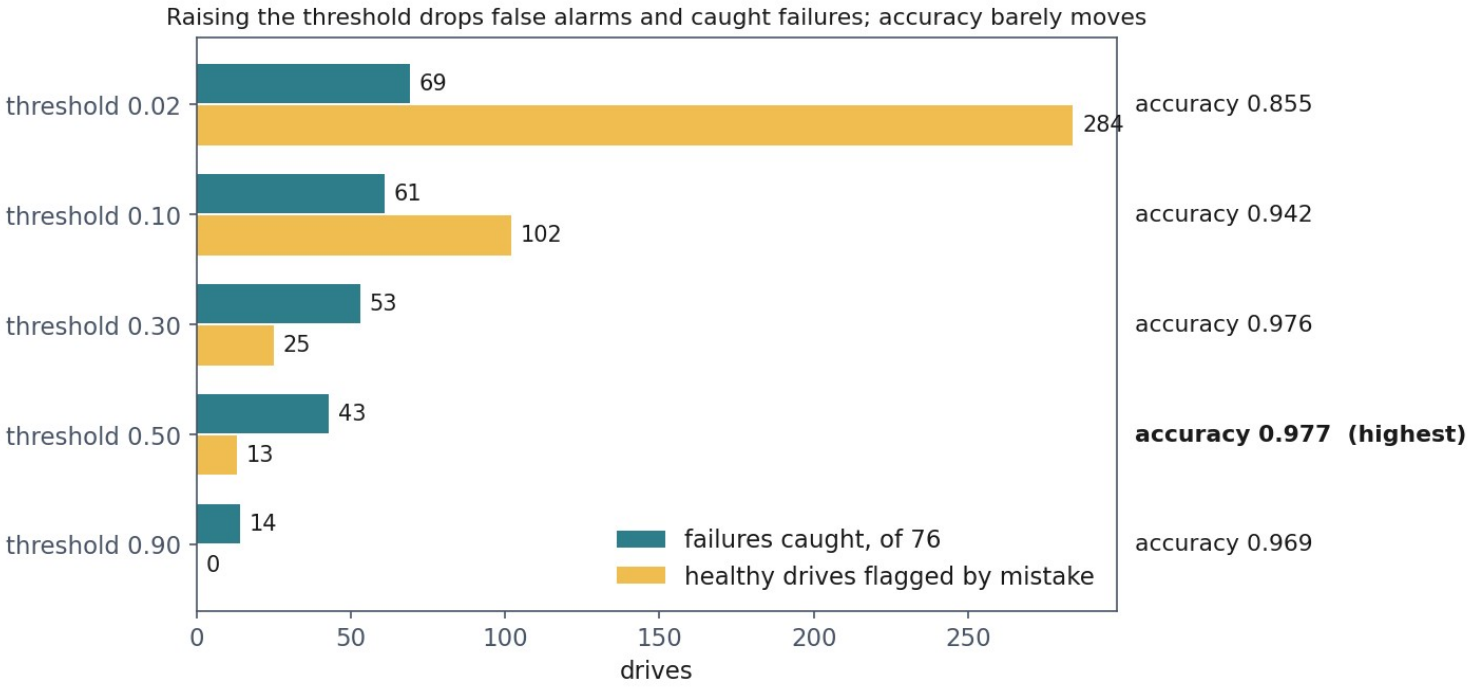
section, and its slides	you call (→ the number to come back with)
the number that should worry you, 31	`accuracy_score` → 96.20% for “always healthy”
the confusion matrix, 33	counts by hand, then `confusion_matrix`, `ConfusionMatrixDisplay` → TP 43
precision and recall, 35-38	`precision_score`, `recall_score`, `f1_score`, `classification_report`
the threshold sweep, 40-41	`predict_proba`, then the metrics at each threshold
the precision-recall curve, 42	`precision_recall_curve`, `average_precision_score` → 0.717 against 0.038

The threshold is a choice nobody made



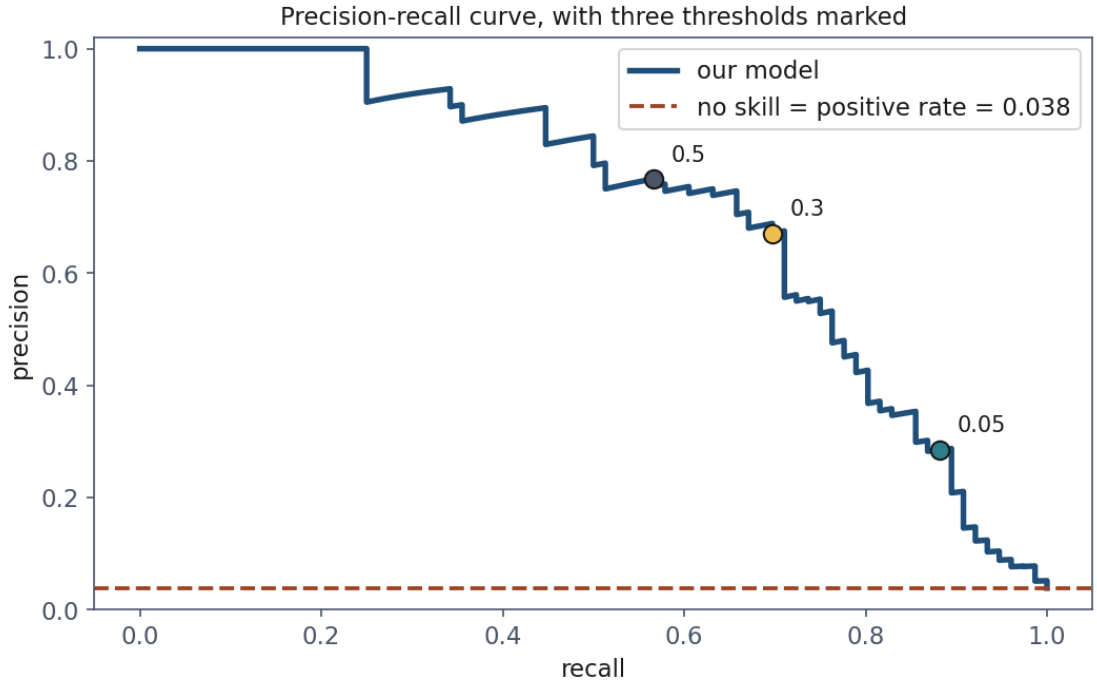
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What the sweep actually costs



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The precision-recall curve



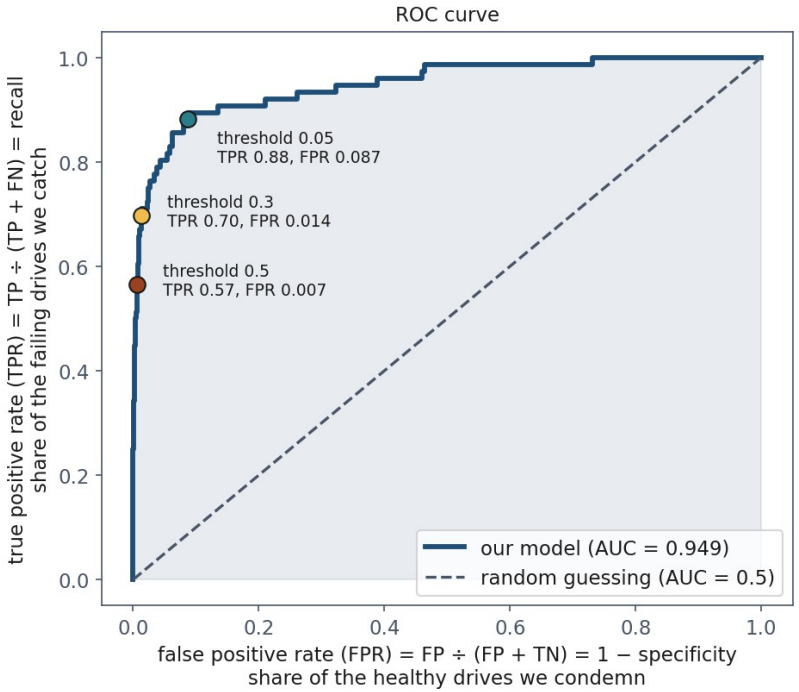
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Comparing models without choosing a threshold

- We keep evaluating at one threshold, but which?
- The **ranking**: the 2,000 test drives sorted by predicted probability, riskiest first
- A threshold is a cut in this list: every drive above the cut is flagged
- What if we score the whole ranking, every cut at once?

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The ROC curve



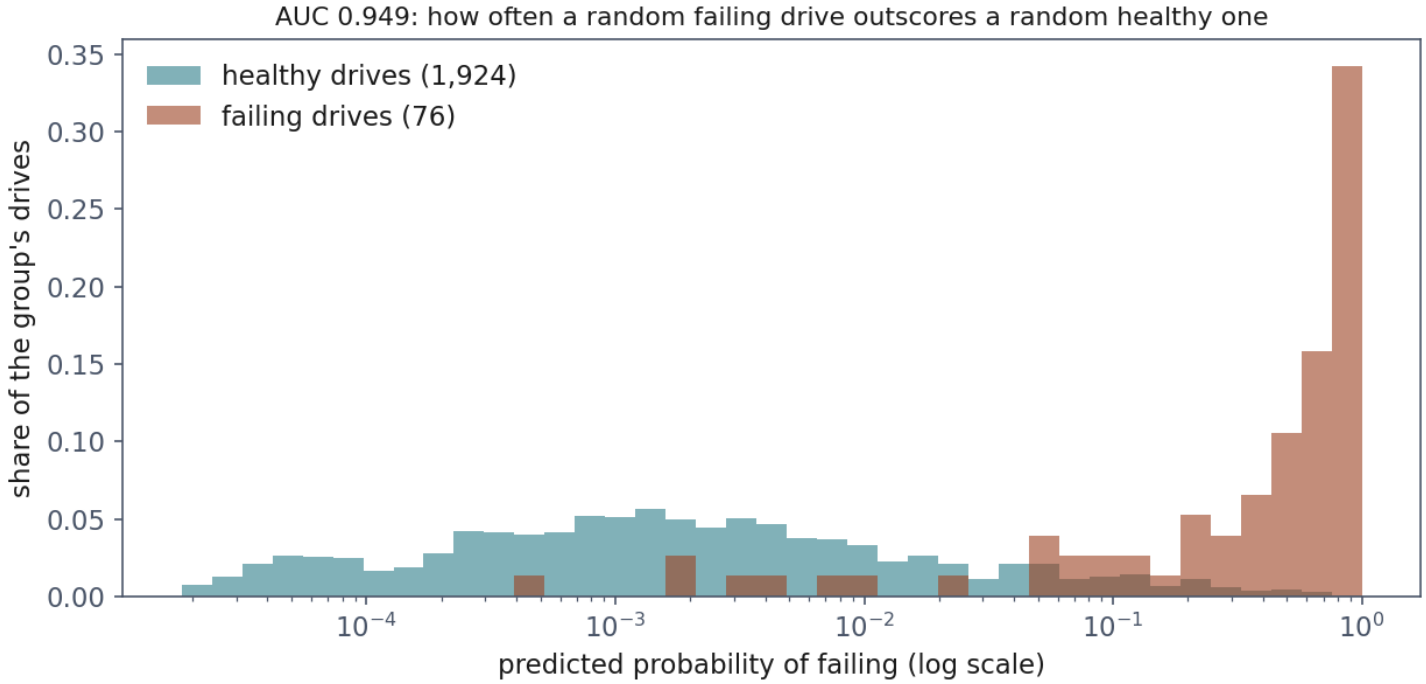
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What the area means

- **Area under the curve (AUC) = the probability that a random failing drive scores above a random healthy one**
- Ours: 0.949 (`roc_auc_score`, notebook 3)
- Check: in 200,000 random pairs, the failing drive scores higher 94.9% of the time
- In code: `roc_auc_score(y_test, scores)`

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The scores behind the AUC



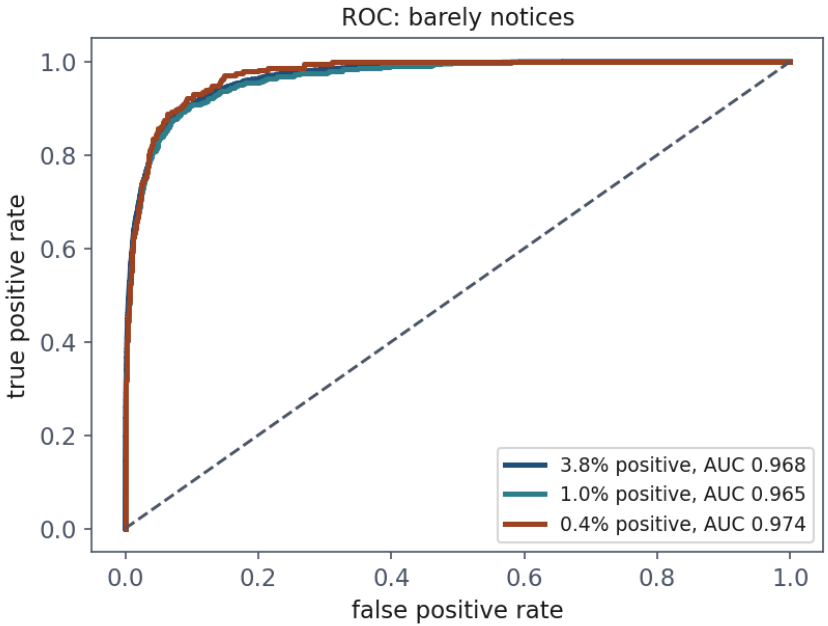
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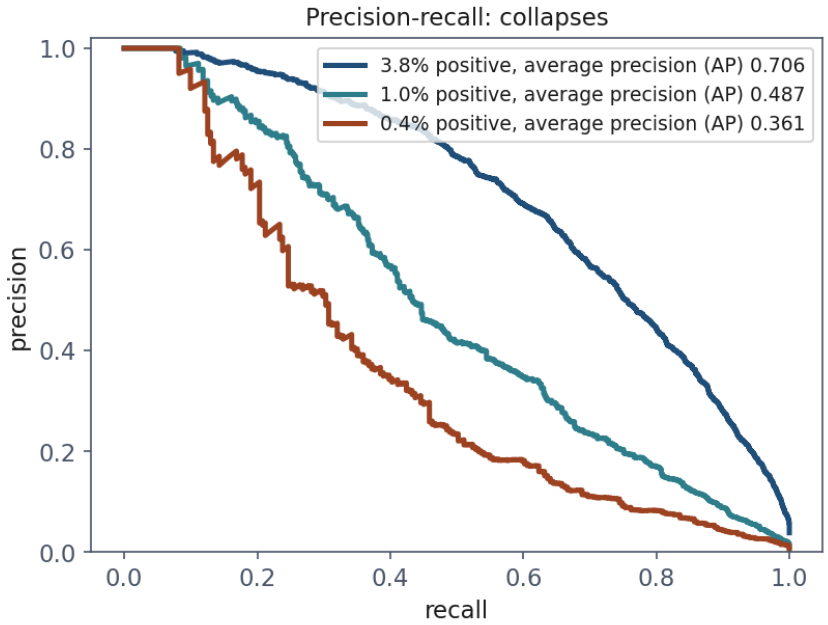
What AUC does not tell you

- Divide every predicted probability by 10
- AUC is **unchanged**, 0.949: the order of the drives is the same
- Every decision changes: at threshold 0.5, 56 drives flagged before, none after (notebook 3)
- The model is now badly **calibrated**: calibrated means a predicted 0.7 comes true about 70% of the time

Where ROC misleads

The same model, not retrained, on 60,000 new drives: every healthy drive kept, failures thinned at random





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Why the two curves disagree

- Recall, shared by both curves: TP ÷ (TP + FN); TP and FN are both failures, thinned in the same proportion: **stays put**
- False positive rate (FPR), the ROC's: FP ÷ (FP + TN), over **all 57,575 healthy drives**, untouched: 0.0053
- Precision, the other curve's: TP ÷ (TP + FP); FP **stays 304**, TP falls from 1,127 to 106

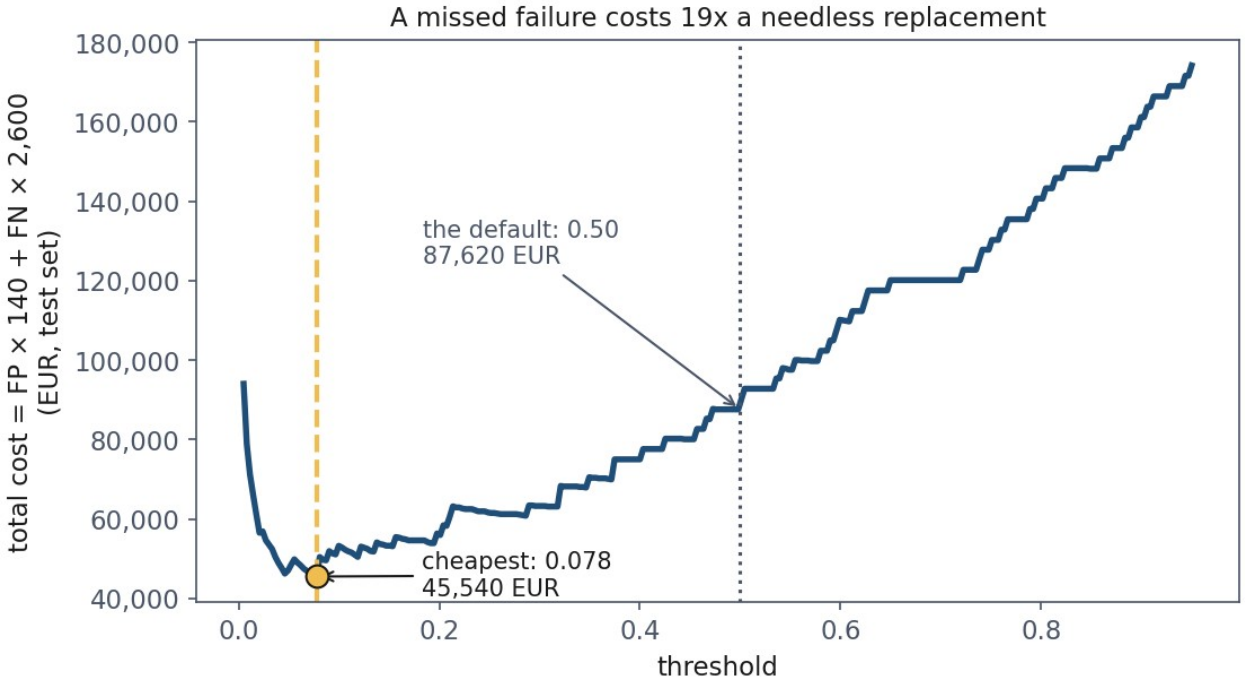
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The rule: report both

- AUC: are failures ranked first? Precision-recall: how many alarms are real?
- On imbalanced data report the **precision-recall curve beside the AUC**
- **Average precision (AP)**: the precision at each failure's place in the ranking, averaged; $\Sigma_k (R_k - R_{k-1}) \times P_k$
- Ours: 0.717 (notebook 2); a random ranking: 76 ÷ 2,000 = 0.038

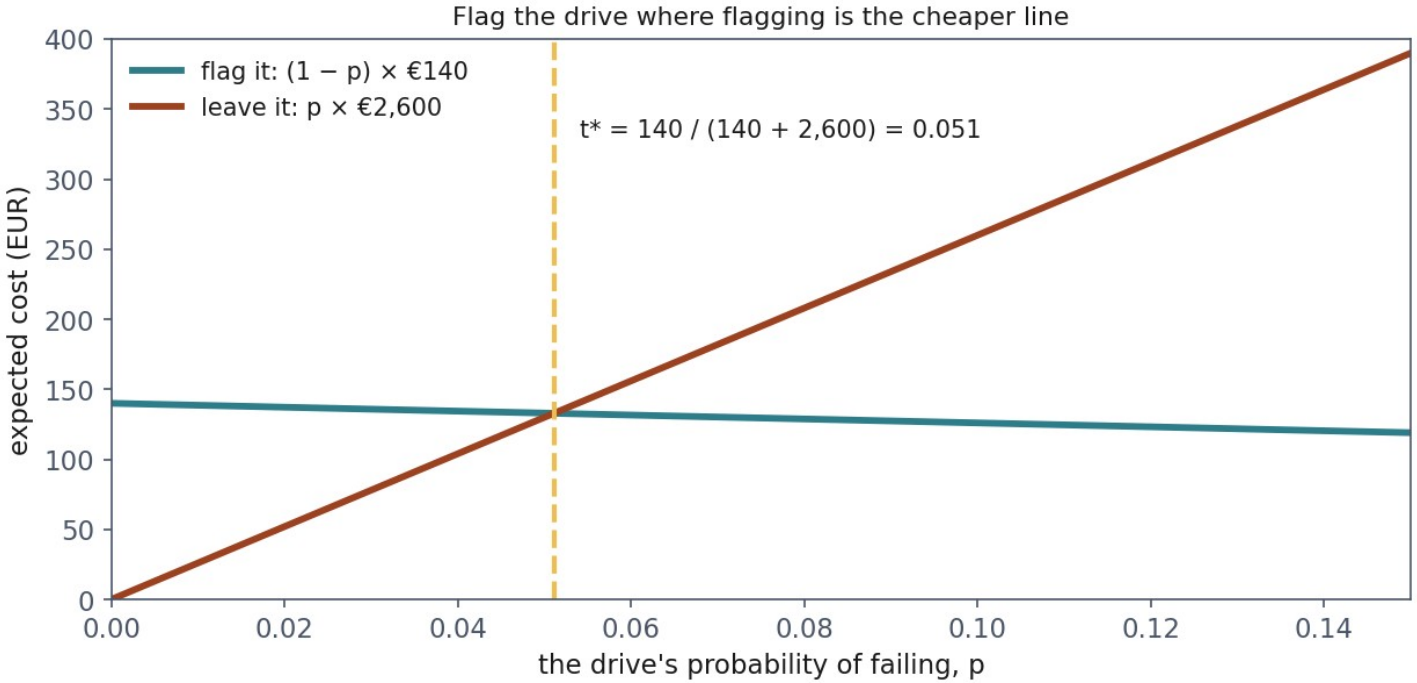
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Choosing the threshold from money



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Where the two costs cross



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Where they cross, in one line

flag if $p\, C_{FN} > (1 - p)\, C_{FP}$

$$\Leftrightarrow \; p > \frac{C_{FP}}{C_{FP} + C_{FN}} = \frac{140}{2740} = 0.051 = t^*$$

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The caveat you must state

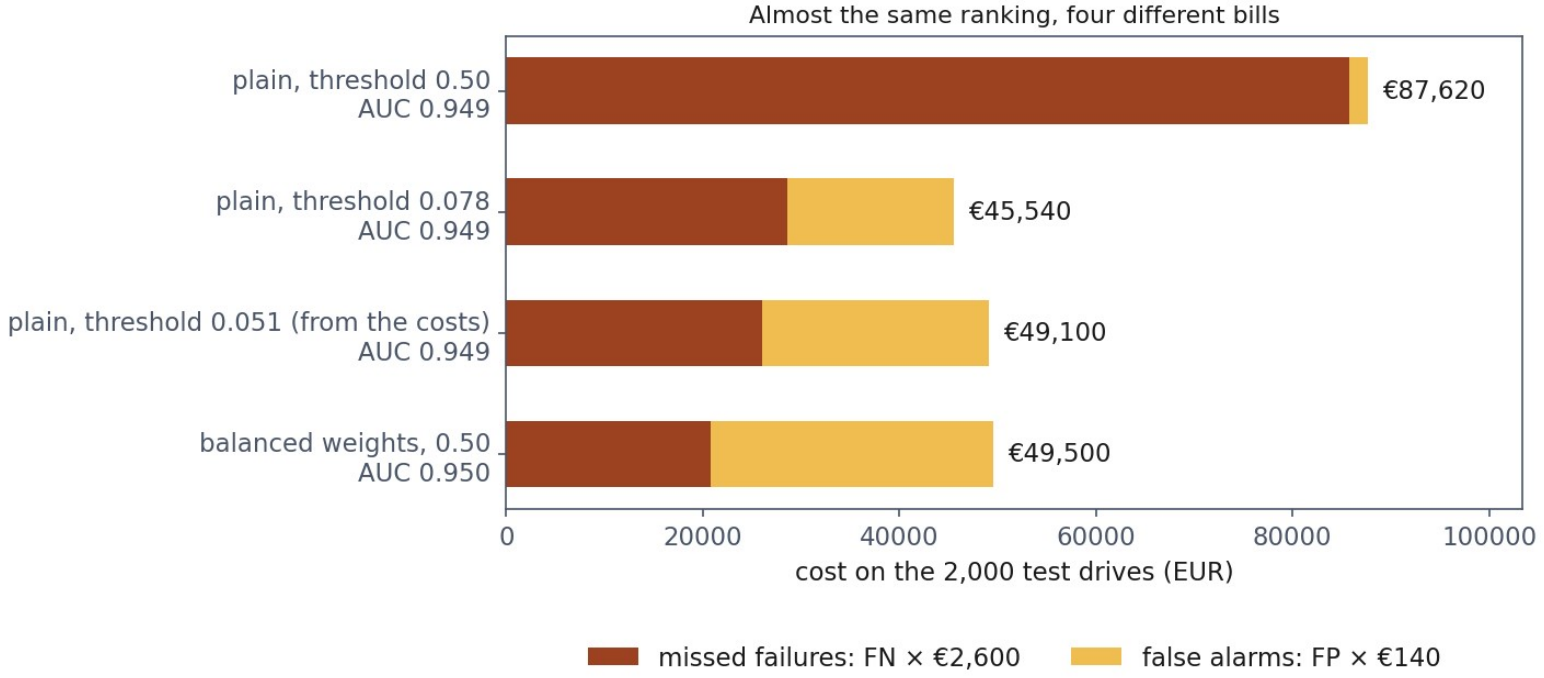
- We chose the threshold **by looking at the test set**: what lesson 1 forbade
- Properly: choose it on a **validation set**, measure once on the test set (lesson 5)
- t* = 0.051, fixed before looking: €49,100 on the test set; 0.078, picked by looking: €45,540
- **Deploy 0.051**: 0.078’s €3,560 was found on the very drives it is measured on

What class weights do

- Training multiplies each drive’s log-loss term by its **class weight**
- `class_weight="balanced"`: m ÷ (2 × class size), 0.52 healthy, 13.0 failed
- A failure counts as 25 healthy drives: **class frequencies**, not costs
- **In effect it only lowers the threshold**: at 0.5 the weighted model decides like the plain one at 0.038

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Class weights: the same lever, pulled earlier



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Notebook 3, live

section, and its slides	you call (→ the number to come back with)
the ROC curve by hand, 44	`roc_curve`, `roc_auc_score` → 0.9493
what AUC means, 45-46	200,000 random (failing, healthy) pairs → 0.9488
the rare-positive trap, 48-49	`average_precision_score` beside `roc_auc_score` → AP 0.706 to 0.361, AUC unmoved
the threshold from costs, 51-54	`confusion_matrix` at each threshold → 45,540 EUR at 0.078, against 87,620 at 0.50
class weights, 55-56	`LogisticRegression` with balanced class weights → AUC 0.950
three classes, 58-60	`classification_report` → macro F1 0.707, weighted 0.883

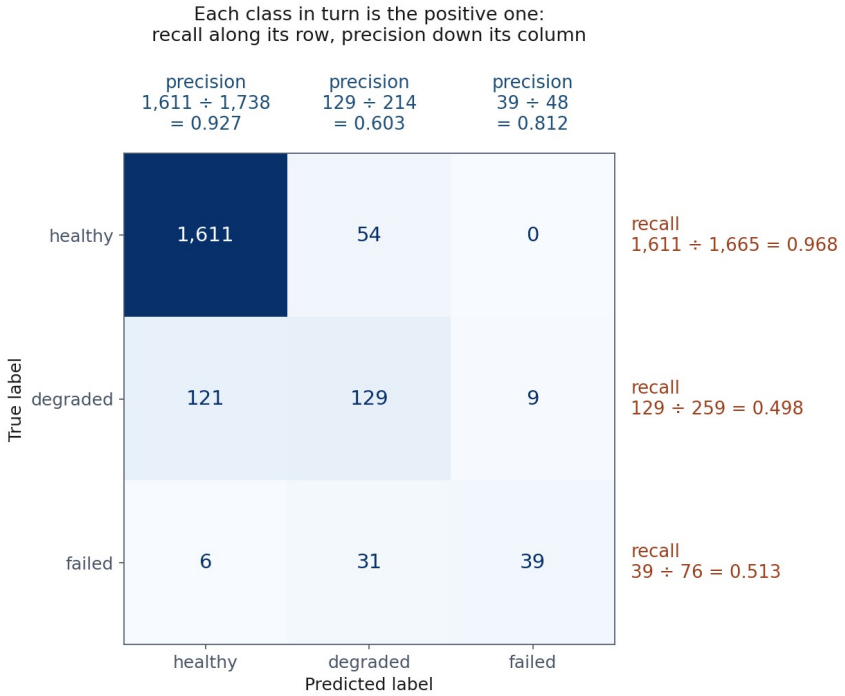
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Three classes: softmax (optional)

- One score per class (healthy, degraded, failed), $z_k = w_k^T x + b_k$; **softmax**: $P(\text{class } k) = \exp(z_k) \div \Sigma_j \exp(z_j)$: exponentiate, divide by the sum
- A degraded drive: scores 0.615, 0.603, −1.218 → 0.466, 0.460, 0.074, called healthy, wrongly
- Predicted class = the largest probability, no 0.5 threshold; two classes: the sigmoid of $z_1 - z_0$
- In code: `decision_function` (scores), `predict_proba` (softmax)

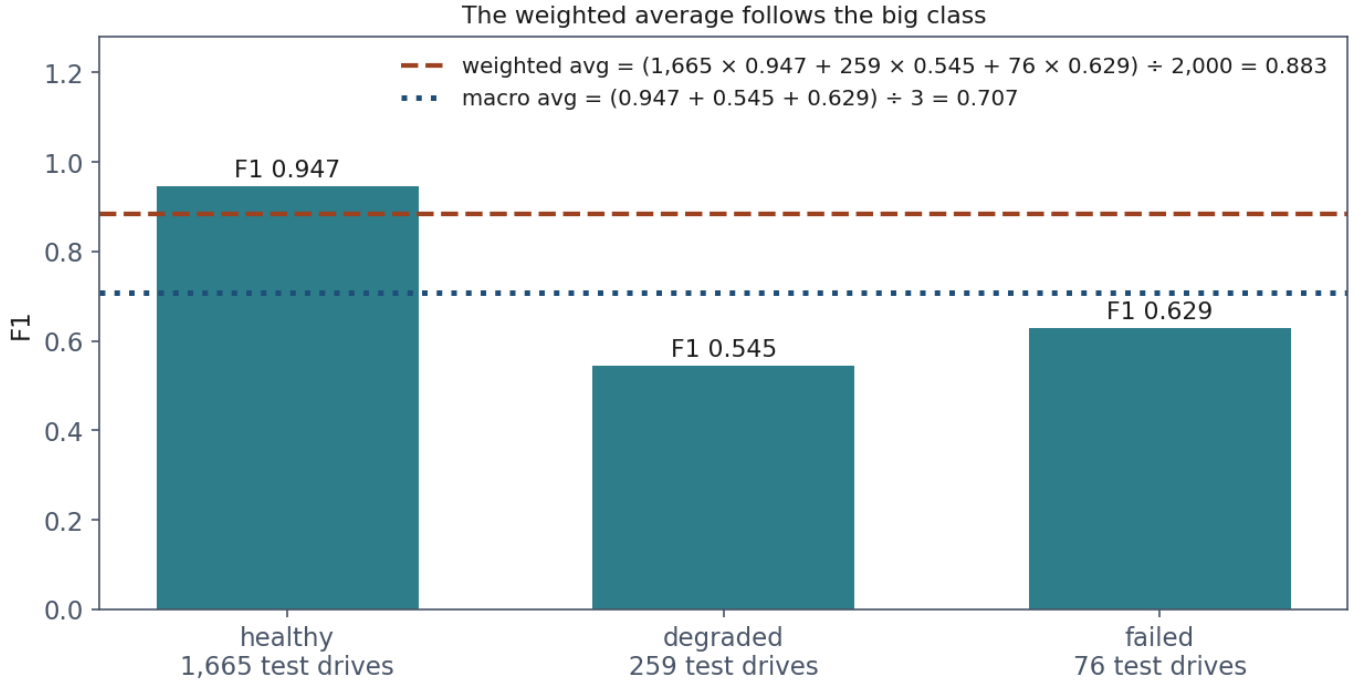
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More than two classes (optional)



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Macro against weighted, one last time (optional)



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What to take away

- Logistic regression: sigmoid of a linear score, fitted by log loss
- **96.20% accuracy, nothing caught**: check the baseline, 1,924 ÷ 2,000
- Precision: the **alarms**; recall: the **failures**
- **Threshold** from costs: 140 ÷ 2,740 = 0.051, if calibrated
- **AUC = ranking**: blind to calibration, hides imbalance

Homework

- **Exercise 4**, discussed at the start of **Friday 13 November**
- Build a classifier, choose a threshold from a stated cost ratio, defend it